Model/year	Developer/country	Key technological features	Primary role/application	Sig
CL-289 (1990)	Dornier (Germany) and Canadair (Canada)	Jet propulsion, GPS navigation, and terrain-following radar.	Tactical battlefield surveillance and mapping.	Enl aut flig NA for rec
RQ-11 Raven (2003)	AeroVironment/United States	Lightweight electric UAV with digital autopilot and portable design.	Tactical reconnaissance and surveillance.	Der mir and dep mo wa

3.6 Challenges and Limitations of Second-generation Drones

While the second generation of drones marked a major technological leap in the evolution of unmanned aerial systems (UAS), this era also faced several technical, operational, and strategic challenges that constrained performance and mission scope. Despite advancements in navigation, communication, and payload integration, UAVs of this period were still in a transitional phase, bridging the gap between manually operated first-generation systems and the fully autonomous, AI-driven drones of later decades. These challenges underscored the technological limitations of the time and highlighted the need for further innovation in automation, endurance, and system reliability.